

D318

Cloud Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An online grocery app with 80,000 weekly customers runs on two stateless web instances behind a load balancer. Traffic doubles on weekends. The app was rewritten in Go last year. The team wants to handle weekend load without downtime and without hitting an instance-size ceiling. What should it do?

- A. Scale up each instance to the largest size.
- B. Rewrite the app again for better speed.
- C. Scale out by adding instances behind the load balancer.
- D. Double the instance count permanently, including weekdays.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A legacy database cannot run on more than one server. It needs more memory to handle load. Which scaling approach applies?

- A. Autoscaling by instance count
- B. Vertical scaling to a larger instance
- C. Horizontal scaling across more servers
- D. No scaling is possible for legacy systems.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A media company with offices in Austin and Toronto needs to store 400 TB of video files accessed by a web app over HTTPS APIs. The files rarely change and each has tags such as title and rights holder. Which storage type fits best?

- A. Object storage
- B. File storage
- C. Local instance storage
- D. Block storage

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A team must attach a low-latency disk to a VM running a transactional database. Which storage type should it use?

- A.** Archive storage
- B.** File storage
- C.** Object storage
- D.** Block storage

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A payment processor handling \$2 million a day says it can lose no more than 15 minutes of transactions in a disaster and must be back online within 4 hours. It currently takes nightly backups to a second region. Which statement is correct?

- A.** Backups every 4 hours would meet the RPO.
- B.** The 4-hour figure is the RPO.
- C.** Nightly backups meet it because they go to a second region.
- D.** Nightly backups fail the 15-minute RPO; it needs replication or backups at least every 15 minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A system takes a snapshot every 6 hours and fails just before the next snapshot. Restoring takes 2 hours. What is the most data it could lose?

- A.** About 2 hours of data
- B.** None, because snapshots always protect all data.
- C.** About 6 hours of data
- D.** About 8 hours of data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A streaming service with a 20-person platform team wants to release a new recommendation engine to 5% of users first, watch error rates, and then widen the release. The release is planned for a Tuesday. Which strategy is this?

- A. Rolling
- B. Blue-green
- C. Big-bang release to all users at once
- D. Canary

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A cloud engineer at a logistics firm with 1,200 trucks creates a subnet 10.20.8.0/26 for application servers. The firm is migrating from a colocation facility in Dallas. Using standard IPv4 rules, how many usable host addresses does the subnet have?

- A. 62
- B. 64
- C. 254
- D. 26

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A subnet must hold 100 hosts. What is the smallest subnet (longest prefix) that fits, using standard IPv4 rules?

- A. /100
- B. /25
- C. /24
- D. /26

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A healthcare startup with 35 employees stores patient files in cloud object storage and already forces TLS 1.3 for all uploads. An auditor who used to work at a large bank asks how the files are protected if someone gains access to the underlying storage media. What is missing?

- A. A newer TLS version
- B. Nothing; TLS 1.3 protects data wherever it is.
- C. Encryption at rest with keys managed in a key management service
- D. Hashing the files instead of encrypting them

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A reporting VM at an accounting firm runs slowly at month-end. Metrics show CPU at 35%, memory at 50%, network at 20%, and disk IOPS pinned at the volume's 3,000 limit. The firm's CFO wants reports by 9 a.m. What change is most likely to help?

- A. Move to a volume type or size with higher provisioned IOPS.
- B. Add more memory.
- C. Add more vCPUs.
- D. Lower the IOPS limit to 35%.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A university runs a student information database 24 hours a day with steady load for the next three years. The database vendor was founded in 1979. Which pricing model will most likely cost the least?

- A. On-demand pricing
- B. Reserved instances or a committed-use discount
- C. Spot, because it is always the cheapest option.
- D. Spot instances

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

An online grocery app ~~with 80,000 weekly customers~~ runs on two stateless web instances behind a load balancer. Traffic doubles on weekends. ~~The app was rewritten in Go last year.~~ The team wants to handle weekend load without downtime and without hitting an instance-size ceiling. What should it do?

Target word: scaling approach **Decides:** Stateless app behind a load balancer; avoid ceilings and downtime.

A	Confusion set	Vertical scaling hits a ceiling and often needs a restart.
B	Camouflage	The language is not the constraint.
C	Correct	Horizontal scaling adds capacity without restarts or size limits.
D	Dropped element	Ignores that load drops on weekdays; wastes money.

2. Answer B Concept 1 • transfer

A legacy database cannot run on more than one server. It needs more memory to handle load. Which scaling approach applies?

Target word: scaling when scale-out is not possible **Decides:** Single-node workload needs more resources on that node.

A	Adjacent category	Autoscaling adds instances, which the app cannot use.
B	Correct	The only way to add memory to a single-node workload.
C	Inverted operation	The database cannot run on more than one server.
D	Absolute language	Vertical scaling still works.

3. Answer A Concept 2

A media company ~~with offices in Austin and Toronto~~ needs to store 400 TB of video files accessed by a web app over HTTPS APIs. The files rarely change and each has tags such as title and rights holder. Which storage type fits best?

Target word: storage type **Decides:** Large unstructured files, API access, metadata tags.

A	Correct	Stores objects with metadata, accessed over HTTP, and scales to petabytes.
B	Adjacent category	Shared folders over NFS or SMB, not an HTTP API with tags.
C	Inverted operation	Temporary and lost when the instance stops.
D	Confusion set	A raw disk for one VM, not API access at this scale.

4. Answer D Concept 2 • transfer

A team must attach a low-latency disk to a VM running a transactional database. Which storage type should it use?

Target word: storage for a database disk **Decides:** A database needs a raw volume attached to a VM.

A	Confusion set	Built for rarely accessed data with slow retrieval.
B	Adjacent category	Network shares add latency and are not typical for database data files.
C	Inverted operation	Accessed by API, not mounted as a database disk.
D	Correct	Block volumes act as attached disks with low latency.

5. Answer D Concept 3

A payment processor ~~handling \$2 million a day~~ says it can lose no more than 15 minutes of transactions in a disaster and must be back online within 4 hours. It currently takes nightly backups ~~to a second region~~. Which statement is correct?

Target word: whether backups meet the RPO **Decides:** RPO is 15 minutes; backups are nightly.

A	Wrong stem number	Uses the RTO; 4-hour backups still risk up to 4 hours of loss.
B	Confusion set	4 hours is the RTO, the allowed downtime.
C	Camouflage	Location helps with regional disasters but not with backup frequency.
D	Correct	Data loss can reach almost 24 hours with nightly backups.

6. Answer C Concept 3 · transfer

A system takes a snapshot every 6 hours and fails just before the next snapshot. Restoring takes 2 hours. What is the most data it could lose?

Target word: maximum data loss from snapshot interval **Decides:** Data loss is set by the time since the last snapshot.

A	Inverted operation	2 hours is the restore time (RTO side), not data loss.
B	Absolute language	Changes after the last snapshot are not protected.
C	Correct	Everything since the last snapshot, up to 6 hours, is lost.
D	Adjacent category	Adds downtime to data loss.

7. Answer D Concept 4

A streaming service ~~with a 20-person platform team~~ wants to release a new recommendation engine to 5% of users first, watch error rates, and then widen the release. ~~The release is planned for a Tuesday.~~ Which strategy is this?

Target word: deployment strategy **Decides:** A small share of traffic goes to the new version first.

A	Adjacent category	Replaces instances in batches rather than targeting a user percentage.
B	Confusion set	Switches all traffic at once between two environments.
C	Inverted operation	The opposite of a gradual release.
D	Correct	A small percentage tests the new version before full rollout.

8. Answer A Concept 5

A cloud engineer at a logistics firm ~~with 1,200 trucks~~ creates a subnet 10.20.8.0/26 for application servers. ~~The firm is migrating from a colocation facility in Dallas.~~ Using standard IPv4 rules, how many usable host addresses does the subnet have?

Target word: usable hosts in a /26 **Decides:** $2^{(32 - 26)} - 2$.

A	Correct	$2^6 = 64$, minus network and broadcast = 62.
B	Dropped element	Leaves out the two reserved addresses.
C	Confusion set	Usable hosts in a /24.
D	Wrong stem number	The prefix length, not a host count.

9. Answer B Concept 5 • transfer

A subnet must hold 100 hosts. What is the smallest subnet (longest prefix) that fits, using standard IPv4 rules?

Target word: solve for prefix from host count **Decides:** Need $2^n - 2 \geq 100$; $n = 7$ gives 126.

A	Wrong stem number	Uses the host count as a prefix; IPv4 prefixes stop at /32.
B	Correct	$2^7 - 2 = 126$ usable, enough for 100; /26 gives only 62.
C	Adjacent category	Fits, but is not the smallest that does.
D	Inverted operation	Too small: 62 usable hosts.

10. Answer C Concept 6

A healthcare startup with 35 employees stores patient files in cloud object storage and already forces TLS 1.3 for all uploads. An auditor who used to work at a large bank asks how the files are protected if someone gains access to the underlying storage media. What is missing?

Target word: which encryption state is missing **Decides:** TLS covers transit; stored data needs encryption at rest.

A	Adjacent category	Still in-transit protection only.
B	Absolute language	TLS protects data only while it moves.
C	Correct	Protects stored data on the media.
D	Confusion set	Hashing checks integrity; the files could not be read back.

11. Answer A Concept 7

A reporting VM at an accounting firm runs slowly at month-end. Metrics show CPU at 35%, memory at 50%, network at 20%, and disk IOPS pinned at the volume's 3,000 limit. The firm's CFO wants reports by 9 a.m. What change is most likely to help?

Target word: fix the saturated resource **Decides:** Disk IOPS is the only resource at its limit.

A	Correct	Removes the actual bottleneck.
B	Adjacent category	Memory at 50% is not saturated.
C	Confusion set	CPU is only at 35%; it is not the constraint.
D	Inverted operation	Tightens the bottleneck instead of relieving it.

12. Answer B Concept 8

A university runs a student information database 24 hours a day with steady load for the next three years. The database vendor was founded in 1979. Which pricing model will most likely cost the least?

Target word: pricing for steady long-term use **Decides:** Always on, predictable, multi-year.

A	Confusion set	Flexible but costs more for steady long-term use.
B	Correct	Commitments give the deepest discount for steady use.
C	Absolute language	Interruptions make it unsuitable, whatever the price.
D	Adjacent category	Cheapest per hour but can be reclaimed; wrong for a critical database.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Horizontal versus vertical scaling	Vertical scaling (up) adds CPU or memory to one instance and has a ceiling and often downtime; horizontal scaling (out) adds instances behind a load balancer and suits stateless apps.	Calling adding instances vertical, or assuming vertical scaling never needs a restart.	Bigger instance versus more instances.
2	Block, file and object storage	Block storage acts as a raw disk for a VM or database; file storage offers shared folders over NFS or SMB; object storage holds files as objects with metadata over HTTP APIs and scales for unstructured data.	Putting a transactional database on object storage.	Who mounts it and how it is accessed: a disk, a share, or an API.
3	RPO and RTO	Recovery point objective is the maximum acceptable data loss measured in time; recovery time objective is the maximum acceptable downtime; backup frequency must be at least as often as the RPO.	Swapping RPO and RTO.	"How much data can we lose" versus "how long can we be down".
4	Deployment strategies	Blue-green switches all traffic between two full environments for instant rollback; canary sends a small share of traffic to the new version first; rolling updates instances in batches.	Calling a small-percentage release blue-green.	How traffic moves to the new version and how rollback works.
5	Subnet sizing	Usable hosts in a subnet = $2^{(32 - \text{prefix})} - 2$ in standard IPv4 networking (network and broadcast addresses reserved).	Forgetting to subtract the reserved addresses, or counting in the wrong direction.	A CIDR prefix and a host count.
6	Encryption at rest versus in transit	At rest protects stored data (disk, database, object keys managed in a KMS); in transit protects data moving over networks (TLS, VPN, IPsec).	Thinking TLS protects data sitting on a disk.	Whether the data is stored or moving.
7	Finding the performance bottleneck	Use metrics to find the saturated resource first (CPU, memory, disk IOPS, network) and fix that resource; adding a resource that is not saturated does not help.	Adding CPU when the metric showing saturation is disk I/O.	A metrics table with one resource near its limit.
8	Cloud pricing models	Reserved or committed-use pricing is cheapest for steady, predictable workloads; spot is cheapest for interruptible jobs; on-demand fits short or unpredictable use.	Putting an always-on production database on spot instances.	Workload pattern: steady, bursty, or interruptible.

Course scope (public description): Cloud Applications prepares learners for the CompTIA Cloud+ certification exam. Learners will gain skills in designing cloud infrastructure and services and in recommending cloud security solutions, policies, and procedures. The course will also introduce skills in deploying cloud solutions for storage, networking, and security, and in managing cloud operations with processes, procedures, and improvements. Learners will also gain skills in troubleshooting cloud services issues in networking, security, and performance.